Feature Engineering for Medicinal Leaf Classification: Ethical Analysis, Visualization-Driven Audits, and Practical Guidance

Momin Khizar Uddin Mohammed
Student, Department of Computer Science
University of North Texas, Denton, TX, USA
Email: MominKhizarUddinMohammed@my.unt.edu

Mohammed Ilyas Ahmed
Student, Department of Computer Science
University of North Texas, Denton, TX, USA
Email: MohammedIlyasAhmedMohammedIlyasAhmed@my.unt.edu

***Abstract*—Medicinal plants are integral to healthcare systems worldwide. Manual identification of plant species is time-consuming and error-prone. This paper presents a feature engineering approach for automated classification of medicinal leaves using the Indian Medicinal Leaf Image Dataset. We explore color, texture, and shape-based features, apply dimensionality reduction, and evaluate multiple machine learning models. Ethical considerations such as fairness, transparency, and responsible AI practices are integrated throughout the pipeline. Results indicate that Support Vector Machines outperform Logistic Regression and Random Forest models, though challenges like class imbalance remain significant.**

I. Introduction (Including Ethical Framework)

Medicinal plants are widely used in traditional and modern medicine, and accurate identification is essential for research and healthcare applications. Automated classification using computer vision can reduce human error and improve efficiency.

From an ethical perspective, our approach emphasizes fairness, transparency, and responsible AI practices. We ensure that the dataset is used in compliance with open-source guidelines and that models do not introduce bias toward specific plant species. Interpretability is prioritized to maintain trust in AI-driven decisions, and the project avoids unnecessary complexity to ensure accessibility for resource-constrained environments. We document preprocessing and modeling choices to support reproducibility and auditability across deployments.

II. Background

The Indian Medicinal Leaf Image Dataset contains 5,224 high-resolution images across 80 distinct classes. Each image represents a unique medicinal plant species, exhibiting variations in color, texture, and shape.

Key challenges include:

- **Class Imbalance:** Some classes have fewer than 5 samples, while others exceed 140.
- **Lighting Variations:** Uneven illumination affects color consistency and can bias color descriptors.
- **Background Noise:** Non-uniform backgrounds complicate segmentation and shape estimation.

A. Related Work

Prior studies emphasize the utility of handcrafted features for plant classification and leaf disease detection. Shape and texture features have been shown to improve classification accuracy; texture methods such as the Gray-Level Co-occurrence Matrix (GLCM) often perform strongly in leaf analysis. Interpretability-focused feature extraction frameworks have highlighted the value of transparent descriptors in plant identification. Hybrid feature engineering strategies that combine complementary cues (e.g., color- and texture-based descriptors) can further improve performance. Recent work also demonstrates competitive results using GLCM and Local Binary Patterns (LBP) in leaf classification tasks [1]–[5].

III. Personal Position

We opted for feature engineering over end-to-end deep learning to achieve interpretability and computational efficiency. While convolutional neural networks (CNNs) offer high accuracy, they often act as black boxes. Our approach allows us to understand which features—color histograms, texture descriptors, or shape metrics—contribute most to classification performance. This transparency aligns with ethical AI principles and supports deployment in resource-constrained environments (e.g., low-power edge devices). Feature engineering also eases domain transfer by enabling targeted adjustments to descriptors without retraining large models.

IV. Data Visualization Implementation

A. Preprocessing and Exploration

Images were resized to 224×224 pixels and normalized. Data augmentation techniques (horizontal/vertical flips, rotations up to 0.2 radians, and zoom up to 0.2) were applied to improve diversity and reduce overfitting.

We computed label distributions for both training (4,180 images) and validation (1,044 images) splits and audited imbalance across classes. The audits confirmed severe skew, including single-digit counts for several classes and over 100 images for others.

B. Feature Engineering

We extracted three categories of features:

- **Color Features:** Average RGB values and per-channel histograms (binned), capturing dominant hues and distribution spread.
- **Texture Features:** GLCM properties (contrast, dissimilarity, homogeneity, energy, correlation, ASM) computed at multiple orientations to capture spatial regularities of leaf surfaces.
- **Shape Features:** Contour-based metrics (area, perimeter, aspect ratio, solidity, eccentricity) inferred via Otsu thresholding and largest-contour selection.

The combined feature vector comprised 56 dimensions per image.

C. Dimensionality Reduction and Imbalance Handling

Principal Component Analysis (PCA) retained 95% variance while reducing feature dimensionality and noise. To mitigate class imbalance, we used SMOTE with a custom sampling strategy that avoids oversampling classes with fewer than two samples, and RandomOverSampler for robustness in alternate settings. Textual audits of resampled label histograms confirmed improved balance while preserving minority classes that could not be safely oversampled.

D. Evaluation Setup

We trained three classifiers on the engineered feature space: Logistic Regression (one-vs-rest), Support Vector Machine (linear kernel), and Balanced Random Forest. Evaluation used accuracy, precision, recall, and F1-score, along with confusion matrices and per-class reports (summarized numerically below).

TABLE I
Model Performance Comparison on Validation Set

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.3103	0.3166	0.3103	0.2889
SVM (Linear)	0.3554	0.3728	0.3554	0.3403
Balanced Random Forest	0.1216	0.2000	0.1200	0.1100

E. Findings from Visualization-Driven Audits (Textual)

Textual summaries indicate: (i) augmentation increases apparent intra-class diversity but does not fully correct skew; (ii) PCA reduces redundancy across color and texture descriptors and stabilizes training; (iii) SVM benefits most from the compacted feature space, while Balanced Random Forest struggles with the limited separability of certain rare classes.

V. Rebuttal

Handcrafted features increase interpretability but can limit accuracy relative to CNN-based models, which learn spatial hierarchies end-to-end. Deep models may achieve higher performance on large image datasets and handle complex variations in texture and shape more effectively. However, the cost is reduced transparency and increased computational demand. A hybrid approach is therefore promising: integrate compact CNN embeddings (e.g., from lightweight backbones) with interpretable descriptors, and apply class-weighted objectives or focal loss to address imbalance while retaining explainability. In low-resource deployments, carefully tuned feature engineering may remain preferable.

VI. Conclusion

We implemented an interpretable feature engineering pipeline for medicinal leaf classification, including color, texture, and shape descriptors, PCA-based compaction, and imbalance mitigation via SMOTE/oversampling. SVM achieved the best performance (accuracy 35.54%, F1 34.03%), followed by Logistic Regression, with Balanced Random Forest trailing due to persistent class skew and feature separability limits. Future work includes weighted loss functions, hybrid CNN+feature approaches, improved segmentation to stabilize shape metrics, and deployment strategies tailored to low-power environments. Throughout, we adhered to ethical AI principles of fairness, transparency, and responsible use of data.

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References

[1] S. Reddy and A. Singh, "Plant Leaf Classification Using Shape and Texture Features," *IEEE Access*, 2021.
[2] R. Patel *et al.*, "Comparative Analysis of Texture-Based Features for Leaf Disease Identification," *Springer Nature*, 2022.
[3] Y. Zhang and J. Li, "Feature Extraction Techniques for Plant Identification," *Computers and Electronics in Agriculture*, 2023.
[4] V. Kumar and D. Sharma, "Hybrid Feature Engineering for Image Classification Tasks," *Pattern Recognition Letters (Elsevier)*, 2020.
[5] M. Joshi and S. Khan, "Application of GLCM and LBP Features in Leaf Classification," *ResearchGate*, 2024.